Seminar infinitesimal and prismatic site

1 Overview of the infinitesimal site

1.1 Topological spaces, presheaves and sheaves on them

Let X be a topological space. We have the category $\mathrm{Open}(X)$, where objects are open subsets $U \hookrightarrow [i]X$ and for $U, V \subseteq X$, $\mathrm{Hom}(U,V) = \begin{cases} \{i\} & \text{if } U \subseteq V \\ \varnothing & \text{otherwise} \end{cases}$.

Definition 1. *1. A presheaf F on X is a contravariant functor $F : \mathrm{Open}(X) \to \mathrm{Sets}$.*

2. F is a sheaf on X if it is a presheaf and $\forall U \subseteq X$ open and $\{U_i\}_{i\in I}$ open cover of U, we have an exact sequence

$$F(U) \longrightarrow \prod_{i\in I} F(U_i) \rightrightarrows \prod_{(i,j)\in I^2} F(U_i \cap U_j)$$

Given an abelian sheaf on X, $X \in \mathrm{Ob}(\mathrm{Open}(X))$ is a final object, i.e. for every $U \in \mathrm{Ob}(\mathrm{Open}(X))$, there exists a unique $U \hookrightarrow X$. Then we can define global sections as $\Gamma(X,F)$ and $H^i(X,F) := R^i\Gamma(X,-)(F)$, with $R^i\Gamma(X,-)$ the ith right derived functor of $\Gamma(X,-)$.

Grothendieck: To define presheaves, sheaves and sheaf cohomology, we only need $(\mathcal{C}, \mathrm{Cov}(\mathcal{C}))$ satisfying certain axioms.

Definition 2. *A site is a pair $(\mathcal{C}, \mathrm{Cov}(\mathcal{C}))$ satisfying that:*

- *$\mathcal{C}$ a category with fibred products.*
- *$\mathrm{Cov}(\mathcal{C})$ a collection, for every $M \in \mathrm{Ob}(\mathcal{C})$ of morphisms $f_i : M_i \to M$ satisfying that*
  - *If $\{M_i \to M\}_{i\in I}$ is a cover of M and $\{M_{ij} \to M_i\}_{J_i}$ are covers of M_i, then $\{M_{ij} \to M\}$ is a cover of M.*
  - *$\mathrm{id} : M \to M$ is a cover of M*
  - *If $M' \to M$, and $M_i \to M$ are a cover, then $M_i \times_M M' \to M'$ is a cover of M'.*

The objects $M \in \mathrm{Ob}(\mathcal{C})$ play the role of open sets, the morphisms $M \to M'$ play the role of inclusions of open sets, and the coverings play the role of open covers.

A presheaf on $(\mathcal{C}, \mathrm{Cov}(\mathcal{C}))$ is a contravariant functor $\mathcal{C} \to \mathrm{Sets}$, and a presheaf is a sheaf if for every $M \in \mathrm{Ob}(\mathcal{C})$, and $\{M_i \to M\}_{i\in I} \in \mathrm{Cov}(M)$, we have that the following diagram is exact:

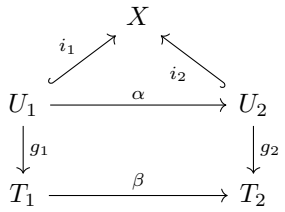
$$F(M) \longrightarrow \prod_{i\in I} F(M_i) \rightrightarrows \prod_{(i,j)\in I^2} F(M_i \cap M_j)$$

1.2 The infinitesimal site

Let K be a field of characteristic 0, and let X be a scheme (or a rigid space/adic space, in this case we must have $K/\mathbb{Q}_p$ finite extension. Suppose that X is smooth over K.

Definition 3. *The infinitesimal site of X consists of:*

- *As a category, we take $\mathcal{C}$ consisting of:*
  - *Objects: Pairs (U,T) where $i : U \hookrightarrow X$ is an open immersion (if we work with rigid spaces it should be an admissible open), and $g : U \to T$ is a nilpotent closed immersion, that is it is a closed immersion of ideal $\mathcal{I} \subseteq \mathcal{O}_T$ and, $\forall t \in T$, there exists T' neighborhood of t such that $(\mathcal{I}|_{T'})^n = 0$ for some n. Observe that if T is quasi-compact, then this condition is equivalent to $\mathcal{I}^N = 0$ for some N. We think as T as a nilpotent thickening of U.*
  - *Morphisms: $(U_1, T_1) \to (U_2, T_2)$ consists of a pair (α, β) with $\alpha : U_1 \to U_2$ open immersion and $\beta : T_1 \to T_2$ is a morphism making the following diagram commutative:*



- *Coverings of an object (U,T) consist of a family $\{(\alpha_i, \beta_i) : (U_i, T_i) \to (U,T)\}$ such that $\{\alpha_i : U_i \to U\}$ is a open covering of U and $\{\beta_i : T_i \to T\}$ is a open covering of T.*

Thus defined, $\mathcal{C}$ is a category with fibred products and equalizers, and $(\mathcal{C}, \mathrm{Cov}(\mathcal{C}))$ is a site, denoted $(X/K)_{\mathrm{inf}}$.

Let X_{inf} denote the infinitesimal topos, that is, the category of sheaves of sets on $(X/K)_{\mathrm{inf}}$. This is the important category (according to Grothendieck), because different sites might give the same topos, so teh topos is the most natural category to consider.

Theorem 1. *1. The data of a presheaf F on $(X/K)_{\mathrm{inf}}$ is equivalent to the data of, for every $(U,T) \in \mathrm{Ob}(\mathcal{C})$, a presheaf $F_{(U,T)}$ on T (for the usual topology, i.e. Zariski in the case of schemes), satisfying that, if we have a morphism $(\alpha, \beta) : (U_1, T_1) to (U_2, T_2)$, then there is a morphism $\phi_{(\alpha,\beta)} : \beta^{-1} F_{(U_2,T_2)} \to F_{(U_1,T_1)}$.*

2. A presheaf is a sheaf on $(X/K)_{\mathrm{inf}}$ if and only if for every (U,T), $F_{(U,T)}$ is a sheaf.

Example 1. *$\mathcal{O}_{\mathrm{inf}}$ the structure sheaf of $(X/K)_{\mathrm{inf}}$, corresponds to the sheaf $\mathcal{O}_{\mathrm{inf}(U,T)} = \mathcal{O}_T$.*

Now we want to discuss global sections. There is a problem, which is that, in general, the category $\mathcal{C}$ that we have defined doesn't have a final object. To solve this, instead of working with a final object in the site, we work with a final object in the topos.

Let $\mathbb{1}^{\mathrm{pre}}(U,T) := \{*\}$ and $\mathbb{1}$ be the sheaffification of the presheaf $\mathbb{1}^{\mathrm{pre}}$. Observe that, if F is a sheaf on $(X/K)_{\mathrm{inf}}$, we have a canonical map $F \to \mathbb{1}^{\mathrm{pre}}$ which is simply given by $F(U,T) \to \mathbb{1}^{\mathrm{pre}}(U,T) = \{*\}$. Now by the formalism of sheaffification, we get a unique map $F \to \mathbb{1}^{\mathrm{pre}} \to \mathbb{1}$. Then, if F is a sheaf on $(X/K)_{\mathrm{inf}}$, we define global sections as $\Gamma((X/K_{\mathrm{inf}}), F) := \mathrm{Hom}_{X_{\mathrm{inf}}}(\mathbb{1}, F)$. Is this correct? It seems to go in the other direction!. Then, we define cohomology as $H^i((X/K)_{\mathrm{inf}}, -) := R^i\Gamma((X/K_{\mathrm{inf}}, -))$

We also remark that there is an explicit description of $\Gamma(X/K_{\mathrm{inf}}, F)$ using envelopes of closed immersions.

Why do all this? Let's shift to (apparently) annother topic.

Definition 4. *On X/K, a module with a connection is a pair $(\mathcal{F}, \nabla)$ where $\mathcal{F}$ is an $\mathcal{O}_X$-module and $\nabla : \mathcal{F} \to \mathcal{F} \otimes_{\mathcal{O}_X} \Omega^1_{X/K}$ is a K-linear map satisfying $\forall U \subseteq X$ open, $a \in \mathcal{O}_X(U)$ and $b \in \mathcal{F}(U)$, $\nabla(ab) = a\nabla(b) + b \otimes da$.*

Given a connection, it induces a sequence

$$\mathcal{F} \underset{\nabla}{\to} \mathcal{F} \otimes_{\mathcal{O}_X} \Omega^1_{X/K} \underset{\nabla^1}{\longrightarrow} \mathcal{F} \otimes_{\mathcal{O}_X} \Omega^2_{X/K} \to \dots$$

via $\nabla^1(b \otimes \omega) = \nabla(b) \wedge \omega + b \otimes d\omega$, for $b \in \mathcal{F}(U)$ and $\omega \in \Omega^1_{X/K}$. Similarly we define ∇^2, ∇^3 and so on.

Definition 5. *We say that a connection ∇ is integrable if $\nabla^1 \circ \nabla = 0$.*

If ∇ is integrable, then the sequence above is a complex, denoted $\mathcal{F} \otimes_{\mathcal{O}_X} \Omega^\bullet_{X/K}$.

Definition 6. *Let X/K be smooth. Then, we define the de Rham cohomology as*

$$H^i_{\mathrm{dR}}(X, (\mathcal{F}, \nabla)) := \mathbb{H}^i(X, \mathcal{F} \otimes_{\mathcal{O}_X} \Omega^\bullet_{X/K})$$

Theorem 2. *Let X/K be smooth, $\mathcal{F}$ be a locally free $\mathcal{O}_X$-module, and $(\mathcal{F}, \nabla)$ be an integrable connection. Then, we have:*

1. *There exists a unique sheaf F_{inf} on $(X/K)_{\mathrm{inf}}$ associated to $(\mathcal{F}, \nabla)$.*
2. *$H^i_{\mathrm{dR}}(X, (\mathcal{F}, \nabla)) \cong H^i((X/K)_{\mathrm{inf}}, F_{\mathrm{inf}})$.*

This works as follows: Given $(\mathcal{F}, \nabla)$ on X, and $(U,T) \in \mathrm{Ob}(\mathcal{C})$, $g : U \to T$ and $T' \geq T$, we have a lift by definition of smoothness $\phi_{T'} : T' \to U$. Then we can define $F_{T'} = \phi^*_{T'}\mathcal{F}$. In general these sheaves don't glue, but the connection allows to glue them and form a sheaf $(F_{\mathrm{inf}})_{(U,T)}$.

Remark 1. *The de Rham cohomology comes with a filtration. The construction of de Rham cohomology using the infinitesimal site does not give the filtration if we work over K. However it can be recovered working in the more general setting of B_{dR}-coefficients.*

Example 2. *If X is smooth and proper, the diagonal embedding $\Delta : X \to X \times_K X$ is a closed immersion, and it factors through the nth infinitesimal neighborhood of the diagonal $\mathcal{P}^n_{X/K}$, for every n. We have that $(X, \mathcal{P}^n_{X/K}) \in \mathrm{Ob}(\mathcal{C})$. In the affine case, $X = \operatorname{Spec} B$, $X \to X \times_K X$ corresponds*

to the multiplication map $B \otimes_K B \to B$ given by $b \otimes b' \mapsto bb'$. Let I be the kernel of this map. Then $\mathcal{P}^n_{X/K} = \mathrm{Spec}(B \otimes_K B/I^{n+1})$, and the first order thickening is corresponds to the sequence

$$0 \to I/I^2 \to B \otimes_K B/I^2 \to B \to 0$$

This should give us an idea of why we can compute de Rham cohomology using that site.

2 Differential operators

Definition 7. *Let X be a topological space, and $\mathcal{O}_1 \to \mathcal{O}_2$ sheaves of rings on X. Let $\mathcal{F}, \mathcal{G}$ be $\mathcal{O}_2$-modules and $k \geq 0$. A differential operator of order $\leq k$, $D \in \mathrm{Diff}^k_{\mathcal{O}_2/s\mathcal{O}_1}(\mathcal{F}, \mathcal{G})$ is a $\mathcal{O}_1$-linear map $D : \mathcal{F} \to \mathcal{G}$ that satisfies the following property. For open $U \subseteq X$, and $a \in \mathcal{O}_2(U)$, the map*

$$s \mapsto D(as) - aD(s)$$

is a differential operator of order $\leq k-1$. We define a differential operator of order 0 to be a $\mathcal{O}_2$-linear map.

Observation 1.

1. *$\mathrm{Diff}^k_{\mathcal{O}_2/\mathcal{O}_1}(\mathcal{F}, \mathcal{G})$ form a $\mathcal{O}_2$-module. This is clear doing induction on k and observing that for $k = 0$ these are $\mathrm{Hom}_{\mathcal{O}_2}(\mathcal{F}, \mathcal{G})$ which is a $\mathcal{O}_2$-module.*
2. *If $n \geq m$, we have $\mathrm{Diff}^n_{\mathcal{O}_2/\mathcal{O}_1}(\mathcal{F}, \mathcal{G}) \supseteq \mathrm{Diff}^m_{\mathcal{O}_2/\mathcal{O}_1}(\mathcal{F}, \mathcal{G})$. Also clear from induction and the fact that the zero map is linear (so a linear map is a differential operator of order ≤ 1).*

Now let's see what this recursive definition means in concrete terms, doing the cases $k = 1$ and $k = 2$.

k=1 Let $s \in \mathcal{F}(U)$ and $a, b \in \mathcal{O}_2(U)$. Then a differential operator of order 1 satisfies that $s \mapsto D(as) - aD(s)$ is linear, so we have $D(abs) - aD(bs) = b(D(as) - aD(s))$, which rearanging terms gives

$$D(abs) = aD(bs) + bD(as) - abD(s)$$

k=2 Let $s \in \mathcal{F}(U)$ and $a, b, c \in \mathcal{O}_2(U)$. Then a differential operator of order 2 satisfies that $D_a := D(as) - aD(s)$ is a differential operator of order 1, so we have, using what we have just obtained above, that $D_a(bcs) = bD_a(cs) + cD_a(bs) - bcD_a(s)$ and substituting the expression of D_a we get $D(abcs) - aD(bcs) = b(D(acs - aD(cs))) + c(D(abs) - aD(bs)) - bc(D(as) - aD(s))$ and rearanging terms:

$$D(abcs) = aD(bcs) + bD(acs) + cD(abs) - abD(cs) - bcD(as) - acD(bs) + abcD(s)$$

Example 3. *If we let $\mathcal{F} = \mathcal{O}_2$, then we can let $s = 1$ in the formula for $k = 1$ obtained above, and we end up with*

$$D(ab) = aD(b) + bD(a) - abD(1)$$

Now define a new map $d : \mathcal{O}_2 \to \mathcal{G}$ by $d(a) = D(a) - aD(1)$, and we have that $d(ab) = D(ab) - abD(1) = aD(b) + bD(a) - abD(1) - abD(1) = a(D(b) - bD(1)) + b(D(a) - aD(1))$. So $d(ab) = ad(b) + bd(a)$ and d is a derivation. Now $D(1) \in \mathcal{G}(U)$, and D is completely determined by d and $D(1)$, so we have

$$\mathrm{Diff}^1(\mathcal{O}_2, \mathcal{G}) = \mathrm{Der}_{\mathcal{O}_2/\mathcal{O}_1}(\mathcal{G}) \oplus \mathcal{G}$$

From now on, we specialize to the situation of schemes: Let $f : X \to S$ be a morphism of schemes that we will assume to be separated. Then we have two sheaves of rings on X, $f^{-1}(\mathcal{O}_S) \to \mathcal{O}_X$. We want to give an equivalent but more geometric notion of what is a differential operator. The rest of this section follows [**?**].

Observe that given a $f^{-1}(\mathcal{O}_X)$-linear map $D : \mathcal{F} \to \mathcal{G}$ we can make it $\mathcal{O}_X$-linear it by setting $\overline{D} : \mathcal{O}_X \otimes_{f^{-1}(\mathcal{O}_S)} \mathcal{F} \to \mathcal{G}$. Now via the isomorphism $\mathcal{F} \cong \mathcal{O}_X \otimes_{\mathcal{O}_X} \mathcal{F}$, we have $\mathcal{O}_X \otimes_{f^{-1}(\mathcal{O}_S)} \mathcal{F} \cong \mathcal{O}_X \otimes_{f^{-1}(\mathcal{O}_S)} \mathcal{O}_X \otimes_{\mathcal{O}_X} \mathcal{F}$, so $\overline{D}$ can be seen as the map $\mathcal{O}_X \otimes_{f^{-1}(\mathcal{O}_S)} \mathcal{O}_X \otimes_{\mathcal{O}_X} \mathcal{F} \to \mathcal{G}$ that is given on sections by $\overline{D}(a \otimes b \otimes s) = aD(bs)$.

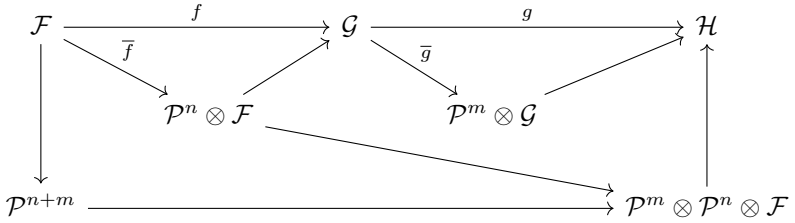
Notation 1. *From now on denote* $\mathcal{P} = \mathcal{O}_X \otimes_{f^{-1}(\mathcal{O}_S)} \mathcal{O}_X$ *and* $\mathcal{I} = \ker(\mathcal{P} \xrightarrow{m} \mathcal{O}_X)$ *where* m *is the multiplication map* $a \otimes b \mapsto ab$*. In other words,* $\mathcal{I}$ *is the ideal defining the diagonal immersion* $X \to X \times_S X$*. We will also denote* $\mathcal{P}^n = \mathcal{P}/\mathcal{I}^{k+1}$*.*

Proposition 1. $D \in \mathrm{Diff}^k_{X/K}(\mathcal{F}, \mathcal{G})$ *if and only if* $\overline{D}$ *factors through* $\mathcal{P} \otimes \mathcal{F} \to \mathcal{P}^k \otimes \mathcal{F} \to \mathcal{G}$*.*

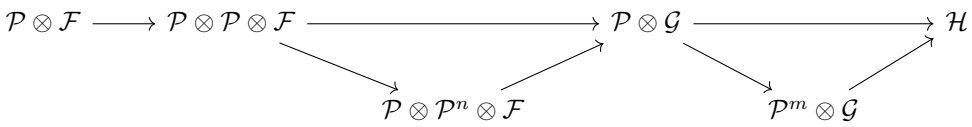
Proof. We proceed by induction on k. For $k = 0$, observe that D is $\mathcal{O}_X$-linear if and only if $sD(x) = D(ax)$ for every $a \in \mathcal{O}_X(U)$, which in terms of $\overline{D}$ is equivalent to $\overline{D}(a \otimes 1 \otimes x) = \overline{D}(1 \otimes a \otimes x)$, or equivalently, $\overline{D}((a \otimes 1 - 1 \otimes a) \otimes x) = 0$. Then as $\mathcal{I}$ is generated by elements of the form $a \otimes 1 - 1 \otimes a$ this is equivalent to $\overline{D}$ factoring through $\mathcal{P}^1$. Now assume that the statement is true for $k-1$. Then $D \in \mathrm{Diff}^k_{X/S}(\mathcal{F}, \mathcal{G})$ if and only if $\forall a$, $D_a \in \mathrm{Diff}^{k-1}(\mathcal{F}, \mathcal{G})$ with $D_a(x) = D(ax) - aD(x)$. But then by induction hypothesis this means that $\overline{D_a}$ factors through $\mathcal{P}^{k-1}$, that is $\overline{D_a}(t \otimes x) = 0$, for every $t \in I^k$. But this is equivalent to $\overline{D}((1 \otimes a)t \otimes x - (a \otimes 1)t \otimes x) = 0$ and we're done as every element of $\mathcal{I}^{k+1}$ can be written as $(a \otimes 1 - 1 \otimes a)t$ for $t \in \mathcal{I}^k$. □

Remark 2.
1. *Proposition 1 tells us that a differential operator can be regarded as a* $f^{-1}(\mathcal{O}_S)$*-linear map that is "almost* $\mathcal{O}_X$*-linear".*
2. *We have* $\mathrm{Diff}^k_{X/S}(\mathcal{F}, \mathcal{G}) \cong \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{P}^k \otimes \mathcal{F}, \mathcal{G})$

Proposition 2. *Let* $f \in \mathrm{Diff}^n_{X/K}(\mathcal{F}, \mathcal{G})$ *and* $g \in \mathrm{Diff}^m_{X/K}(\mathcal{G}, \mathcal{H})$*. Then* $g \circ f \in \mathrm{Diff}^{n+m}_{X/K}(\mathcal{F}, \mathcal{H})$ *and we have a commutative diagram*



Proof. Observe that the only thing that we need to prove is that the diagram commutes, as then $\overline{g \circ f}$ does indeed factor through $\mathcal{P}^{n+m} \otimes \mathcal{F}$ and so and therefore we have $g \circ f \in \mathrm{Diff}^{n+m}_{X/S}(\mathcal{F}, \mathcal{G})$. Then observe that we have $\overline{g \circ f} = \overline{g} \circ (\mathrm{id}_{\mathcal{O}_X} \otimes_{f^{-1}(\mathcal{O}_S)} f)$ as both map $a \otimes b \otimes s \mapsto a(g \circ f)(bs)$. Now we express $(\mathrm{id}_{\mathcal{O}_X} \otimes_{f^{-1}(\mathcal{O}_S)} f)$ in terms of $\overline{f}$: For that, consider the map $\delta : \mathcal{P} \to \mathcal{P} \otimes_{\mathcal{O}_X} \mathcal{P}$ given on sections by $a \otimes b \mapsto a \otimes 1 \otimes 1 \otimes b$. This is a ring homomorphism. Then consider the composition $\overline{\delta}(f) : \mathcal{P} \otimes \mathcal{F} \xrightarrow[\delta \otimes \mathrm{id}_{\mathcal{F}}]{} \mathcal{P} \otimes_{\mathcal{O}_X} \mathcal{P} \otimes_{\mathcal{O}_X} \mathcal{F} \xrightarrow[\mathrm{id}_{\mathcal{P}} \otimes \overline{f}]{} \mathcal{G}$. Via the isomorphism $\mathcal{O}_X \otimes_{f^{-1}(\mathcal{O}_S)} \mathcal{F} \cong \mathcal{P} \otimes_{\mathcal{O}_X} \mathcal{F}$ we have $\overline{\delta}(f) = \mathrm{id}_{\mathcal{O}_X} \otimes_{f^{-1}(\mathcal{O}_S)} f$, as we can check on sections: $\overline{\delta}(f)(a \otimes b \otimes s) = (\mathrm{id}_{\mathcal{P}} \otimes \overline{f})(a \otimes 1 \otimes 1 \otimes b \otimes s) = (a \otimes 1) \otimes f(bs)$. Therefore it follows that $\overline{g} \circ \overline{\delta}(f) = \overline{g \circ f}$. This gives us a diagram



which tells us that $\overline{g \circ f}$ factors through $\mathcal{P}^m \otimes \mathcal{P}^n \otimes \mathcal{F}$. Therefore, the only thing we need to check is that $\delta_{m,n} : \mathcal{P} \xrightarrow[\delta]{} \mathcal{P} \otimes \mathcal{P} \to \mathcal{P}^m \otimes \mathcal{P}^n$ factors through $\mathcal{P}^{m+n}$. For that, observe that $\mathcal{I}$ is

generated by elements $\eta = 1 \otimes x - x \otimes 1$, and $\delta(\eta) = 1 \otimes 1 \otimes 1 \otimes x - x \otimes 1 \otimes 1 \otimes 1$. Now the tensor product in the middle is with respect to $\mathcal{O}_X$, so $1 \otimes x \otimes 1 \otimes 1 = 1 \otimes 1 \otimes x \otimes 1$, and so we can write $\delta(\eta) = \mathbb{1} \otimes \eta + \eta \otimes \mathbb{1}$, where $\mathbb{1} = 1 \otimes_{f^{-1}(\mathcal{O}_S)} 1$. Now as δ is a ring homomorphism, every generator of $\mathcal{I}^{n+m+1}$ will be sent to $\prod_{i=1}^{n+m+1}(\mathbb{1} \otimes \eta_i + \eta_i \otimes \mathbb{1})$ and every term when we expand the product will have either $m+1$ η's in the first $\mathcal{P}$-coordinate or $n+1$ in the second, so it will vanish modulo $\mathcal{I}^{m+1} \otimes \mathcal{I}^{n+1}$. □

Proposition 3. *Let X/S be smooth and $x_1, \ldots, x_n$ be local coordinates, that is, we have an étale map $g : X \to \mathbb{A}^n_S$ defined by the sections $x_1, \ldots, x_n \in \mathcal{O}_X(X)$. Denote $\eta_i = 1 \otimes x_i - x_i \otimes 1$. Then*

1. *$\mathcal{P}^n$ is a free $\mathcal{O}_X$-module with basis the images of $\{\eta_1^{\alpha_1} \cdots \eta_n^{\alpha_n}$ such that $\sum \alpha_i \leq n\}$.*
2. *Denote $q = (q_1, \ldots, q_n)$ and let $\overline{D}_q$ be the dual basis of η_q in $\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{P} \otimes \mathcal{F}, \mathcal{G})$. Let D_q be the corresponding differential operators. Then $\{D_q$ for $|q| \leq m\}$ is a basis of $\mathrm{Diff}^m_{X/S}(\mathcal{O}_X, \mathcal{O}_X)$ and the composition is given by*

$$D_q \circ D_{q'} = \binom{q+q'}{q} D_{q+q'}$$

where the facorial of a multiindex is defined by $q! = \prod q_i!$.

Proof. 1. It follows from results on regular sequences. There are proofs of some of the statements about regular sequences used in [**?**]. Recall that for a ring A, and elements $a_i \in A$, $(a_1, \ldots, a_n)$ is a regular sequence if, for every i, a_i is not a zero divisor in $A/(a_1, \ldots, a_{i-1})$. We say that a morphism of schemes $f : X \to Y$ is a regular immersion if $\forall x \in X$, $\mathcal{O}_{Y,f(x)} \to \mathcal{O}_{X,x}$ has kernel generated by a regular sequence. A surjective morphism of local regular rings has kernel generated by a regular sequence. Then, in our situation, the diagonal embedding $X \to X \times_S X$ is closed so the local maps are surjective, and $f : X \to S$ is smooth, so the local rings are regular and therefore $X \to X \times_S X$ is a regular immersion.

Now we have the following lemma:

Lemma 1. *If $\mathcal{I}$ is an ideal of $\mathcal{O}_X$ that is locally generated by a regular sequence, it satisfies that $\mathcal{I}^j/\mathcal{I}^{j+1} \cong \mathrm{Sym}^j(\mathcal{I}/\mathcal{I}^2)$.*

Observe that the étale map g gives an isomorphism $g^*\Omega_{\mathbb{A}^n_S/S} \cong \Omega^1_{X/S}$, and therefore $\Omega^1_{X/S}$ is free generated by the images of η_i. Then by the lemma above $\mathcal{I}^j/\mathcal{I}^{j+1}$ is also free generated by the images of $\eta_1^{\alpha_1} \cdots \eta_n^{\alpha_n}$ with $\sum \alpha_i = j$. Then we can do induction on the sequences (which split because $\mathcal{P}^{j-1}$ is free by induction hypothesis, and therefore projective)

$$0 \to \mathcal{I}^j/\mathcal{I}^{j+1} \to \mathcal{P}^j \to \mathcal{P}^{j-1} \to 0$$

and we conclude that $\mathcal{P}^j$ is free for every j and the elements $\{\eta_1^{\alpha_1} \cdots \eta_n^{\alpha_n}$ such that $\sum \alpha_i \leq n\}$ form a basis.

2. It is clear that the D_q's form a basis of $\mathrm{Diff}^m_{X/S}(\mathcal{O}_X, \mathcal{O}_X)$, so we just need to check the statement about the composition. For that, it's enough to check how $\overline{D_q \circ D_{q'}} = \overline{D_q} \circ (\mathrm{id}_{\mathcal{P}} \otimes$

$\overline{D_{q'}}) \circ \delta$ act on η^p (recall here p is a multiindex). Recall from the proof of Proposition 2 that δ is a ring homomorphism, and so we have

$$\delta(\eta^p) = (\eta \otimes \mathbb{1} + \mathbb{1} \otimes \eta)^p = \sum_{i+j=p} \binom{p}{i} (\eta \otimes \mathbb{1})^i (\mathbb{1} \otimes \eta)^j$$

Now applying $\overline{D_{q'}}$ only the terms with $j = q'$ survive because of the dual basis definition, and the same with $\overline{D_q}$, only the terms with $i = q$ survive, and we end up with the formula of the statement.

□

Remark 3. *(about Proposition 3)*

1. *It seems that this still works even if $X \to S$ is not separated. Then the diagonal map is still a locally closed immersion. However the ideal $\mathcal{I}$ may not be defined in all X, only locally, but $\mathcal{I}^j/\mathcal{I}^{j+1}$ can be defined in all X because they're naturally $\mathcal{O}_X$-modules.*
2. *The result implies in particular that the D_q's commute.*
3. *If S is a scheme over $\mathbb{Q}$, then we are able to invert the factorials, and so $\mathrm{Diff}^1_{X/S}(\mathcal{O}_X, \mathcal{O}_X)$ generate $\mathrm{Diff}^m_{X/S}(\mathcal{O}_X, \mathcal{O}_X)0$ for every m.*

Example 4. *Let $X = \mathbb{A}^2_K$, for K a field of characteristic 0. Then, every differential operator $D \in \mathrm{Diff}^k_{\mathbb{A}^2_K/K}(\mathcal{O}_X, \mathcal{O}_X)$ is of the form*

$$D = \sum_{m+n \leq k} \frac{\partial^{m+n}}{\partial x^n \partial y^m}$$

To see that, given the result of Proposition 3, it is enough to see what are D_0, D_x and D_y, the differential operators corresponding to the dual basis of $1, x \otimes 1 - 1 \otimes x, y \otimes 1 - 1 \otimes x$. For that it is enough to see what they do on monomials $x^n y^m$.

- *$D_0(1) = 1$, and $0 = \overline{D_0}(1 \otimes x - x \otimes 1) = D_0(x) - xD_0(1)$, so $D_0(x) = x$ and similarly $D_0(y) = y$. Now by induction it's easy to see that $D_0(x^m y^n) = x^n y^m$. Assume for instance $n \geq 1$, and that the result holds for $n' + m' < n + m$. Then as it is a order 1 differential operator $0 = \overline{D_0}(x^{n-1}y^m \otimes 1 - 1 \otimes x^{n-1}y^m)(x \otimes 1 - 1 \otimes x)) = D(x^n y^m) - xD_0(x^{n-1}y^m) - x^{n-1}y^m D_0(x) + x^n y^m D_0(1)$ and so $D_0(x^n y^m) = x^n y^m$.*
- *$D_x(1) = 0$ and $1 = \overline{D_x}(1 \otimes x - x \otimes 1) = xD_x(1) - D_x(x)$ so $D_x(x) = 1$ and the same shows $D_x(y) = 0$. Now by induction again we can show $D_x(x^n y^m) = nx^{n-1}y^m$ for $n \geq 1$ and $D_x(y^m) = 0$. Again by being a order 1 differential operator, we get $0 = \overline{D_x}(x^{n-1}y^m \otimes 1 - 1 \otimes x^{n-1}y^m)(x \otimes 1 - 1 \otimes x)) = D(x^n y^m) - xD_x(x^{n-1}y^m) - x^{n-1}y^m D_x(x) + x^n y^m D_x(1)$ and by induction hypothesis we have $D(x^n y^m) = xD_x(x^{n-1}y^m) + x^{n-1}y^m D_x(x) = x(n-1)x^{n-2}y^m + x^{n-1}y^m = nx^{n-1}y^m$. Therefore $D_x = \frac{\partial}{\partial x}$*

The symmetric reasoning shows $D_y = \frac{\partial}{\partial y}$. In conclusion, every differential operator or order ≤ 1 can be written as

$$D = P(x,y)\frac{\partial}{\partial x} + Q(x,y)\frac{\partial}{\partial y} + R(x,y)$$

and as already said, the more general statement follows because D_x and D_y generate $D_{(n,m)}$ for higher order differential operators.

2.1 Connections as differential operators

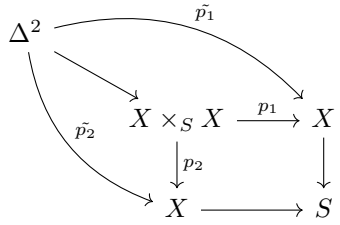
Definition 8. *Let $X \to S$ be a morphism of schemes, and $\mathcal{F}$ a $\mathcal{O}_X$-module. Then, a connection on $\mathcal{F}$ is a map*

$$\nabla : \mathcal{F} \to \mathcal{F} \otimes \Omega^1_{X/S}$$

satisfying, on sections, for every $a \in \mathcal{O}_X(U)$ and $x \in \mathcal{F}(U)$, $\nabla(ax) = a\nabla(x) + x \otimes da$.

Observation 2. *Observe that a connection is in particular a order 1 differential operator. Indeed, $s \mapsto \nabla(as) - a\nabla(s) = s \otimes da$ which is clearly $\mathcal{O}_X$-linear: $bs \mapsto bs \otimes da = b(s \otimes da)$.*

We have a more geometric equivalent definition of connection: Denote Δ^2 the closed subscheme of $X \times_S X$ defined by the ideal $\mathcal{I}^2$, that is a first order thickening of X. Then we have maps



Proposition 4. *A connection $\nabla : \mathcal{F} \to \mathcal{F} \otimes \Omega^1_{X/S}$ is equivalent to a $\mathcal{P}^1_{X/S}$-linear isomorphism $\epsilon : \tilde{p_2}^*\mathcal{F} \to \tilde{p_1}^*\mathcal{F}$.*

Proof. Given ϵ, define $\theta : \mathcal{F} \to \mathcal{F} \otimes \mathcal{P}^1$ by $\theta = \epsilon \circ p_2^{\#}$. Then we have that θ is linear for the right $\mathcal{O}_X$-structure on $\mathcal{P}$, that is, we have $\theta(ax) = (1 \otimes a)\theta(x)$. Now consider ∇ defined by $\nabla(x) = \theta(x) - x \otimes \mathbb{1}$. Now ϵ reduces to id $\mod \mathcal{I}$, which tells us that ∇ has image in $\mathcal{I}/\mathcal{I}^2 = \Omega^1_{X/S} \subseteq \mathcal{P}^1$, so the target is well defined. Now we just need to check that it satisfies the correct condition on sections. But indeed, $\nabla(ax) = (1 \otimes a)\theta(x) - ax \otimes \mathbb{1} = (1 \otimes a)\theta(x) + (1 \otimes a)(x \otimes \mathbb{1}) - (1 \otimes a)(x \otimes \mathbb{1}) - ax \otimes \mathbb{1} = (1 \otimes a)\nabla(x) + da \otimes x$.

Reciprocally, given a connection ∇, let $\theta(x) = \nabla(x) + x \otimes \mathbb{1}$. The same computation shows that θ is linear for the right $\mathcal{O}_X$ structure on $\mathcal{P}^1$. Then tensoring with θ we get a $\mathcal{P}^1$-linear map $\epsilon : \tilde{p_2}^*\mathcal{F} = \mathcal{P}^1 \otimes \mathcal{F} \to \mathcal{F} \otimes \mathcal{P}^1 = \tilde{p_1}^*\mathcal{F}$. and it clearly gives the identity modulo $\mathcal{I}$, by construction.

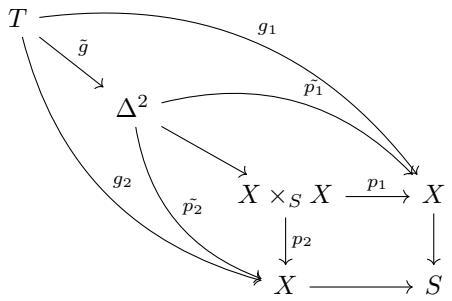
Now consider the involution $\tau : \mathcal{P}^1 \to \mathcal{P}^1$ given by $a \otimes b \mapsto b \otimes a$. This induces, tensoring with $\mathcal{F}$, a τ-linear map $\sigma : p_1^*\mathcal{F} \to p_2^*\mathcal{F}$, and gives a τ-linear map $\sigma \circ \epsilon : p_2^*\mathcal{F} \to p_2^*\mathcal{F}$. Now $(\sigma \circ \epsilon)^2$ is $\mathcal{P}^1$-linear, and reduces to the identity $\mod \mathcal{I}$. Then by Lemma 2 $(\sigma \circ \epsilon)^2$ is an isomorphism, so $\sigma \circ \epsilon$ is bijective and so is ϵ. This shows that ϵ is actually an isomorphism. □

Lemma 2. *Let $\varphi : M \to M$ be a A-linear map that reduces to* id *modulo an ideal $I \subseteq A$ such that $I^2 = 0$. Then φ is an isomorphism.*

Proof. Simply let $h(x) = \varphi(x) - x$. Then $h : M \to IM$. Consider then $\psi = x - h(x)$. We have $\psi \circ \varphi(x) = \psi(x + h(x)) = x + h(x) - h(x + h(x)) = x$, and similarly $\varphi \circ \psi = \mathrm{id}$, so φ is an isomorphism. □

Remark 4. *Proposition 4 tells us that a connection allows us to descend a sheaf on X to the first order thickening of X, Δ^2. That is, via the retractions $\tilde{p_i}^*$, we have in general two ways of pulling back the section $\mathcal{F}$ from X to Δ^2. But if $\mathcal{F}$ has a connection we have a way of identifying these pullbacks, and so a unique way to descend $\mathcal{F}$ to Δ^2 (up to isomorphism).*

More generally, if we have any other first order thickening of X, $X \to T$, and there exists a retraction $g_1 : T \to X$, then we have a unique (up to iso) way of descending $\mathcal{F}$ to T if $\mathcal{F}$ has a connection. Indeed, if there is any other retraction $g_2 : T \to X$, then $g = (g_1, g_2) : T \to X \times_S X$ will factor through Δ^2. Then we can pullback the isomorphism ϵ from Δ^2 to T along $\tilde{g}$ and we have a way of identifying the pullbacks $g_1^*\mathcal{F}$ and $g_2^*\mathcal{F}$.



Example 5. *What is the corresponding map ϵ for the connection given by the exterior derivative $d : \mathcal{O}_X \to \Omega^1_{X/S}$? Well, $d(x) = 1 \otimes x - x \otimes 1$ and so $\theta(x)$ is just the image of $1 \otimes x$ in $\mathcal{P}^1$. Now ϵ is the $\mathcal{P}^1$-linearization of θ, so $\epsilon(a \otimes b) = (a \otimes 1)\epsilon(1 \otimes b) = (a \otimes 1)(1 \otimes b) = a \otimes b$, so ϵ is just the identity map.*

3 Connections

3.1 De Rham Cohomology

Let $\mathcal{A}, \mathcal{B}$ be abelian categories, and $\mathcal{A}$ with enough injectives, $F : \mathcal{A} \to \mathcal{B}$ be a left exact functor. Let $\mathcal{D}(\mathcal{A})$ be the category of bounded from below complexes: Objects in $D(\mathcal{A})$ are complexes

$$M^\bullet : \cdots \to M^i \to M^{i+1} \to \dots$$

such that there exists some n_0 with $M^i = 0 \ \forall i < n_0$. Then F induces a functor $D(F) : \mathcal{D}(\mathcal{A}) \to \mathcal{D}(\mathcal{B})$, and we call hyperderived functors of F the derived functors of $D(F)$. More precisely:

Definition 9. *If $A^\bullet \in \mathcal{D}(\mathcal{A})$, an injective resolucion is a map $0 \to A^\bullet \xrightarrow{\phi} I^\bullet$ such that*

a ϕ is injective.

b I^n is injective in $\mathcal{A}$ for all n

c ϕ is a quasi-isomorphism, that is, $H^n(\phi) : H^n(A^\bullet) \to H^n(I^\bullet)$ is an isomorphism.

We define the hyperderived functors of F as $\mathbb{R}^i F(M^\bullet) := H^i(F(I^\bullet))$.

Remark 5. *The definition of injective resolution in the category $D(\mathcal{A})$ is made such that $\mathbb{R}^n F(M^\bullet)$ is independent, up to canonical isomorphism, of the injective resolution.*

In particular, we are interested in the following situation:

Definition 10. *Let X be a scheme, and let $\mathcal{F}^\bullet : \mathcal{F}^0 \to \mathcal{F}^1 \to \dots$ be a complex of abelian sheaves on X. Then the ith hypercohomology of $\mathcal{F}^\bullet$ is $\mathbb{H}^i(X, \mathcal{F}) := \mathbb{R}^i\Gamma(X, -)(\mathcal{F})$. In particular, for the particular case of the de Rham complex $\Omega_X^\bullet : \mathcal{O}_X \to \Omega_X^1 \to \dots$, we have the de Rham cohomology $H^i_{\mathrm{dR}}(X) = \mathbb{H}^i(X, \Omega_X^\bullet)$.*

Now we'll see a recipie to calculate the de Rham cohomology of a separated, noetherian scheme X, and $\mathcal{F}^\bullet$ in $\mathrm{QCoh}(X)$. We first need a lemma.

Lemma 3. *Let $\varphi : M^\bullet \to N^\bullet$ be a quasi-isomorphism of bounded complexes such that N^k is F-acyclic, i.e. $R^iF(N^k)$, for all $i \geq 1$ and for all k. Then $\mathbb{R}^n F(M^\bullet) \cong H^n F(N^\bullet)$.*

Remark 6. *The content of the above lemma is that we can compute hyperderived functors using acyclic resolutions.*

Proof. Let $M^\bullet \to N^\bullet$ be a quasi-isomorphism. Then we have that $\mathbb{R}^i F(M^\bullet) = \mathbb{R}^i F(N^\bullet)$ (I think this is not immediate, but we showed it in p-adic Hodge theory). So, it suffices to prove that for $M^\bullet$ F-acyclic we have $\mathbb{R}^i F(M^\bullet) \cong H^i(F(M^\bullet))$. Now let $M^\bullet \to I^\bullet$ be an injective resolution and consider the short exact sequence $0 \to M^\bullet \to I^\bullet \to Q^\bullet \to 0$, that is, for every k, we have $0 \to M^k \to I^k \to Q^k \to 0$ exact. Now we take R^iF on this sequence, and we have $\forall i \geq 1$, the exact sequence

$$0 \to F(M^k) \to F(I^k) = 0 \to F(Q^k) \to R^iF(M^k) = 0$$

And therefore this shows that Q^k is F-acyclic for every k. Now taking cohomology we get an exact sequence

$$H^i(N^\bullet) \cong H^i(I^\bullet) \to H^i(Q^\bullet) \to H^{i+1}(M^\bullet) \cong H^{i+1}(I^\bullet)$$

and therefore this shows that $Q^\bullet$ is an exact complex. Now being exact and F-acyclic implies that $F(Q^\bullet)$ is exact as well, and therefore taking the long exact sequence associated in cohomology to the short exact sequence $0 \to F(M^\bullet) \to F(I^\bullet) \to F(Q^\bullet) \to 0$ we get $H^n(F(Q^\bullet)) = 0$ and therefore $H^n(F(M^\bullet)) = H^n(F(I^\bullet)) = \mathbb{R}^n(F(I^\bullet))$. □

Now let's go back to the goal of calculating the de Rham cohomology of a separated noetherian scheme X. Let $\mathcal{F}^\bullet$ be a complex with $\mathcal{F}^i \in \mathrm{QCoh}(X)$. Let $\mathfrak{U} = \{U_i\}_{i\in I}$ be a finite affine cover of X. Recall the Čech complex associated to $\mathcal{F} = \mathcal{F}^i$ for every fixed i.

$$\cdots \to \check{C}^p(\mathfrak{U}, \mathcal{F}) = \prod_{i_0<\cdots<i_p} \mathcal{F}(U_{i_0}\cap\cdots\cap U_{i_p}) \xrightarrow[\delta^p]{} checkC^{p+1}(\mathfrak{U}, \mathcal{F}) = \prod_{i_0<\cdots<i_{p+1}} \mathcal{F}(U_{i_0}\cap\cdots\cap U_{i_{p+1}}) \to \dots$$

with $\delta_p(a)_{i_0,\dots,i_{p+1}} = \sum_{k=0}^{p+1}(-1)^k a_{i_0,\dots,\hat{i_k},\dots i_{p+1}}|_{U_0\cap\cdots\cap U_{p+1}}$. Now let's denote for simplicity $U_{i_0,\dots,i_p} = U_{i_0}\cap\cdots\cap U_{i_p} \hookrightarrow [j_{i_0,\dots,i_p}]X$. Observe that we can sheaffify the Čech complex, defining $\check{\mathcal{C}}^p(\mathfrak{U}, \mathcal{F}) = \prod_{i_0<\cdots<i_p} j_{i_0,\dots,i_p *} j^*_{i_0,\dots,i_p}\mathcal{F}$. This sheaf satisfies $\Gamma(X, \check{\mathcal{C}}^p(\mathfrak{U}, \mathcal{F})) = \check{C}^p(\mathfrak{U}, \mathcal{F})$. Therefore, using that the scheme is separated and that $\mathfrak{U}$ is an open affine cover, we have that

$$H^n(X, \check{\mathcal{C}}^\bullet(\mathfrak{U}, \mathcal{F})) = \bigoplus H^n(U_{i_0,\dots,i_p}, \mathcal{F}|_{i_0,\dots,i_p}) = 0 \quad \text{for } n > 0$$

which means that $\check{\mathcal{C}}^p(\mathfrak{U}, \mathcal{F})$ is Γ-acyclic. Then,

$$0 \to \mathcal{F} \to \check{\mathcal{C}}^p(\mathfrak{U}, \mathcal{F})$$

is exact and is an acyclic resolution for $\mathcal{F}$. In particular, we have $\check{H}^i(\check{\mathcal{C}}^\bullet(\mathfrak{U}, \mathcal{F}) = H^i(\check{C}^\bullet(\mathfrak{U}, \mathcal{F}) = H^i(X, \mathcal{F})$. Now we want to build an acyclic resolution for the complex of sheaves $\mathcal{F}^\bullet$. We consider the double complex:

$$\begin{array}{ccccc}
 & & \vdots & & \vdots \\
 & & \uparrow & & \uparrow \\
\dots & \longrightarrow & E^{p+1,q} = \check{\mathcal{C}}^{p+1}(\mathfrak{U}, \mathcal{F}^q) & \xrightarrow{(-1)^{p+1}d^q} & E^{p+1,q+1} = \check{\mathcal{C}}^{p+1}(\mathfrak{U}, \mathcal{F}^q) \longrightarrow \dots \\
 & & \uparrow{\scriptstyle \delta^p} & & \uparrow{\scriptstyle \delta^p} \\
\dots & \longrightarrow & E^{p,q} = \check{\mathcal{C}}^{p}(\mathfrak{U}, \mathcal{F}^q) & \xrightarrow{(-1)^{p}d^q} & E^{p+1,q+1} = \check{\mathcal{C}}^{p}(\mathfrak{U}, \mathcal{F}^{q+1}) \longrightarrow \dots \\
 & & \uparrow & & \uparrow \\
 & & \vdots & & \vdots
\end{array}$$

Let $\mathrm{T}^\bullet$ be the total complex associated to the double complex above, that is

$$\mathrm{T}^n = \bigoplus_{p+q=n} E^{p,q} \xrightarrow[D]{} \mathrm{T}^{n+1} = \bigoplus_{p+q=n+1} E^{p,q}$$

with $(D^n(a))_{p,q} = \delta^{p-1,q}(a_{p-1,q}) + (-1)^p d^{p,q-1}(a_{p,q-1})$. We have that $F^\bullet \to \mathrm{T}^\bullet$ is an acyclic resolution and $\mathbb{H}^n(X, \mathcal{F}^\bullet) \cong H(\Gamma(X, \mathrm{T}^\bullet))$.

Example 6. *Let X/K be a smooth projective curve, $\Omega^1_{X/K}$ is locally free of rank one, and we have $\Omega^i_{X/K} := \bigwedge^i \Omega^1_{X/K} = 0$ for $i > 1$. Then the de Rham complex is simply $\mathcal{O}_X \to \Omega^1_{X/K}$, and we have*

$H^i_{\mathrm{dR}}(X) \cong H^i(T^\bullet)$, *where* $T^\bullet$ *is the total complex associated to the following Čech double complex: Let* $\mathfrak{U} = \{U, V\}$ *be an open cover by two affines (always possible for a projective curve).*

$$\begin{array}{ccc} \check{\mathcal{C}}^0(\mathfrak{U}, \Omega^1_{X/K}) = \Omega^1_{X/K}(U) \oplus \Omega^1_{X/K}(U) & \xrightarrow{\delta^0} & E^{p+1,q+1} = \check{\mathcal{C}}^1(\mathfrak{U}, \Omega^1_{X/K}) = \Omega^1_{X/K}(U \cap V) \\ \uparrow{\scriptstyle d} & & \uparrow{\scriptstyle -d} \\ \check{\mathcal{C}}^0(\mathfrak{U}, \mathcal{O}_X) = \mathcal{O}_X(U) \oplus \mathcal{O}_X(V) & \xrightarrow{\delta^0} & \check{\mathcal{C}}^p(\mathfrak{U}, \mathcal{O}_X) = \mathcal{O}_X(U \cap V) \end{array}$$

namely, we have

$$T^0 = \mathcal{O}_X(U) \oplus \mathcal{O}_X(V) \xrightarrow[D^0]{} T^1 = \Omega^1_X(U) \oplus \Omega^1_X(V) \oplus \mathcal{O}_X(U \cap V) \xrightarrow{D^1} T^2 = \Omega_X(U \cap V)$$

with $D^0(f_U, f_V) = (dF_U, df_V, f_U|_{U\cap V} - f_V|_{U\cap V})$ *and* $D^1(\omega_U, \omega_V, f_{UV}) = \omega_U|_{U\cap V} - \omega_V|_{U\cap V} - df_{UV}$. *Therefore, we have*

- $H^0_{\mathrm{dR}}(X) = \ker D^0 = \{(f_U, f_V) \in \mathcal{O}_X(U) \oplus \mathcal{O}_X(V) | df_U = 0 = df_V$ *and* $f_U|_{U\cap V} = f_V|_{U\cap V}\} = \mathcal{O}_X(X) \cong K$
- $H^1_{\mathrm{dR}}(X) = \frac{\ker D^1}{\operatorname{Im} D^0} = \frac{(\omega_U, \omega_V, f_{UV}) \in \Omega^1_X(U) \oplus \Omega_X(V) \oplus \mathcal{O}_X(U\cap V) | \omega_U|_{U\cap V} - \omega_V|_{U\cap U} - df_{UV} = 0}{(df_U, df_V, f_U|_{U\cap V} - f_V|_{U\cap V} | (f_U, f_V) \in \mathcal{O}_X(U) \oplus \mathcal{O}_X(V))}$. *It has dimension* $2g$, *where* g *is the genus.*
- $H^2_{\mathrm{dR}}(X) = \frac{\Omega^1_X(U\cap V)}{\operatorname{Im} D^1}$

We can also define a relative version of de Rham cohomology:

Definition 11. *Let* $\pi : X \to S$ *be a smooth morphism of schemes over* K. *Then the relative de Rham cohomology is the sheaf* $H^i_{\mathrm{dR}}(X/S) = \mathbb{R}\pi_* \Omega^\bullet_{X/S}$ *on* S.

3.2 Connections

Connections will allow to have other coefficients for the de Rham cohomology that are not the structure sheaf. Let X be a smooth scheme over K, and $\mathcal{F} \in \mathrm{QCoh}(\mathcal{O}_X)$.

Definition 12. *A connection on* $\mathcal{F}$ *is a* K*-linear morphism* $\nabla : \mathcal{F} \to \Omega^1_X \otimes_{\mathcal{O}_X} \mathcal{F}$ *satisfying on sections* $\nabla(af) = da \otimes f + a\nabla(f)$.

Observe that a connection gives rise to maps $\nabla_i : \Omega^i_{X/K} \otimes \mathcal{F} \to \Omega^{i+1}_X \otimes \mathcal{F}$ defined by $\omega \otimes f \mapsto d\omega \otimes f + (-1)^i \omega \wedge \nabla(f)$.

Definition 13. *Given* $(\mathcal{F}, \nabla)$ *a sheaf with connection, the curvature of* ∇ *is the* $\mathcal{O}_X$*-linear map* $K := \nabla_1 \circ \nabla : \mathcal{F} \to \Omega^2_X \otimes \mathcal{F}$. *One can check that, with this definition,* $()\nabla^{i+1} \circ \nabla^i)(\omega \otimes f) = \omega \wedge K(f)$.

Definition 14. *We say that the connection* ∇ *is integrable if* $K(\nabla) = 0$, *or, equivalently, if* ∇^i *give rise to a complex* $\mathcal{F} \xrightarrow[\nabla]{} \mathcal{F} \otimes \Omega^1_X \xrightarrow[\nabla^1]{} \mathcal{F} \otimes \Omega^2_X \to \ldots$. *In that case, we denote this complex* $\Omega^\bullet_X(\mathcal{F})$, *and we define* $H^i_{\mathrm{dR}}(X, \mathcal{F}, \nabla) = \mathbb{H}^i(X, \Omega^\bullet_X(\mathcal{F}))$.

Remark 7. *A connection gives rise to a* K*-linear map* $\mathrm{Der}(\mathcal{O}_X) \to \mathrm{End}_K(\mathcal{F})$ *given by* $D \mapsto \nabla(D)$, *where* $\nabla(D) : \mathcal{F} \to \mathcal{F}$ *is the map* $\mathcal{F} \xrightarrow[\nabla]{} \mathcal{F} \otimes \Omega^1_X \xrightarrow[\varphi_D \otimes 1]{\mathcal{O}_X} \otimes \mathcal{F}$, *defined via the identification* $D \leftrightarrow \varphi_D$ *given by the isomorphism* $\mathrm{Der}_K(\mathcal{O}_X) \cong \mathrm{Hom}_K(\Omega^1_X, \mathcal{O}_X)$.

We should do this exercise for the Gauss-Manin connection.

Example 7. *1.* $d : \mathcal{O}_X \to \Omega^1_X \cong \Omega^1_X \otimes \mathcal{O}_X$ *is a connection on* X*.*

2. *Let* $\mathcal{F} \cong \mathcal{O}_X$ *and let* $\omega \in H^0(X, \Omega^1_X)$*. Then* $\nabla(1) = \omega$ *defines a connection:* $\nabla(a) = \nabla(a1) = a\nabla(1) + da \otimes 1 \cong a\omega + da$*.*

3. *Let* $\mathcal{F}$ *be free of rank* d *over* $\mathcal{O}_X$*, i.e.* $\mathcal{F} \cong \mathcal{O}_X^d$*, and let* $e_1, \dots, e_d \in H^0(X, \mathcal{F})$ *be generators of* $\mathcal{F}$*. Let* $\omega_1, \dots, \omega_d \in H^0(X, \Omega^1_X)$*. Then,* $\nabla(e_i) := \omega_i$ *defines a connection on* $\mathcal{F}$*. Indeed,* $\forall s \in \mathcal{F}(U) \cong \bigoplus_{i=1}^d \mathcal{O}_X(U)e_i$*,* $s = a_1 e_1|_U + \cdots + a_d e_d|_U$*,* $a_i \in \mathcal{O}_X(U)$*. Then,*

$$\nabla(s) = \sum_{i=1}^d \nabla(a_i e_i|_U) = \sum_{i=1}^d da_i \otimes e_i|_U + a_i \omega_i \otimes e_i|_U$$

4. *Let* $\mathcal{L}$ *be a line bundle on* X*. Then, there exists a (finite, open) cover* $\{U_i\}_{i \in I}$ *that trivializes it:* $\mathcal{L}|_{U_i} \cong \mathcal{O}_X|_{U_i} e_i$*, for* $e_i \in \mathcal{L}(U_i)$*. Then,* $e_i|_{U_i \cap U_j} = a_{ij} e_j|_{U_i \cap U_j}$*, for some* $a_{ij} \in \mathcal{O}_X^\times(U_{ij})$*, and* $a_{ik|_{U_{ijk}}} = a_{ij}|U_{ijk} a_{jk}|_{U_{ijk}}$*. So, this defines a Čech cocycle, and we have a map*

$$\begin{aligned} \mathrm{Pic}(X) \cong H^1(X, \mathcal{O}_X^\times) &\to H^1(X, \Omega^1_X) \\ \mathcal{L} \mapsto [(a_{ij})_{i,j}] &\mapsto \left[\left(\eta_{ij} = \frac{da_{ij}}{a_{ij}} \right)_{i,j} \right] \end{aligned}$$

Let $[\mathcal{L}] = 0$ *on* $H^1(X, \Omega^1_X)$*, which implies* $\exists (\omega_i)_{i \in I}$*, with* $\omega_i \in \Omega^1_X(U_i)$ *such that* $\frac{da_{ij}}{a_{ij}} = \eta_{ij} = \omega_i|_{U_{ij}} - \omega_j|_{U_{ij}}$*. Then we set* $\nabla_i(e_i) = \omega_i \otimes e_i$*, and by the previous example, these define Connections* $\nabla_i : \mathcal{F}|_{U_i} \to \Omega^1_X|_{U_i} \otimes \mathcal{F}|_{U_i}$*. We now check that the connections* ∇_i *glue and therefore define a connection on* $\mathcal{F}$*. Indeed, on intersections we have*

$$\nabla_i(e_j|_{U_{ij}})$$

3.3 Connections over $\mathbb{C}$

For this section we follow [**?**].

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